P510/1
PHYSICS
Paper 1
AUGUST, 2026
2½ hours

JINJA JOINT EXAMINATIONS BOARD

Uganda Advanced Certificate of Education

MOCK EXAMINATIONS – AUGUST, 2026

PHYSICS

Paper 1

2 hours 30 minutes

INSTRUCTIONS TO CANDIDATES:

Attempt only **four** items
Section A is compulsory. Section B has two parts (I) and (II). Attempt not more than one item from each of the Parts (I) and (II).
Use the provided data and supporting materials to guide your solutions.
Clearly show all calculations and assumptions.
Responses to each item should start on a fresh page
Silent non-programmable scientific calculators may be used.

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SECTION A

ITEM 1

Uganda is facing challenges of high electricity costs and frequent power fluctuations. A team from the Ministry of Energy has been tasked with exploring alternative sources of energy to help address these challenges. As part of their investigation, the team visited the Kyambogo University laboratory, where radioactive samples were being studied.

During a trial experiment, 1 kg of uranium-235 ($^{235}_{92}U$)was placed in a nuclear reactor and bombarded with neutrons to initiate nuclear fission. However, the team did not know how much energy would be released during the reaction and therefore could not complete their report. Their target is to produce energy not less than 6.0×10^{13}J

As they were leaving the laboratory, one member slipped on a staircase and was rushed to the hospital. A medical imaging device was used to determine whether any bones had been fractured. On that very day, a group of Senior Six students who had come for a study tour were asked to help the medical team understand how the device works and how bone fractures can be detected; however, they could not provide sufficient information.

A few days later, some members of the team began experiencing symptoms such as skin irritation and burns. After a medical examination, a student pursuing a medical radiography course was selected to give a medical report on what could have caused the symptoms experienced by the team members. He found that during the investigation in the laboratory, the team did not observe adequate safety precautions while handling radioactive samples, thereby exposing themselves to the harmful effects of radiation.

Support information:

$$^{235}_{92}U + {}^{1}_{0}n \rightarrow {}^{148}_{57}La + {}^{85}_{35}Br + 3\,{}^{1}_{0}n + \text{energy}$$

Mass of ^{235}U = 235.1 u
Mass of ^{148}La = 148.0 u
Mass of neutron = 1.009 u
Mass of ^{85}Br = 84.9 u

1U = 931MeV

Task: You have been selected as a Senior Six Physics student to:

(a) Examine the information provided and help the Senior Six students who attended the tour, as well as the radiography student, to compile their report.
(b) Analyse the scenario and help the Ministry of Energy team complete their report by providing the missing information.

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ITEM 2

During the transportation of construction materials to a bridge construction site in a rural district of Uganda, a truck driver noticed that one of the wheel rims became very hot after repeatedly applying the brakes while descending a steep hill. He suspected that the heating was due to overloading of the truck. However, the conductor explained that the heating resulted from the frictional force between the brake pads and the wheel rim. Although his explanation was partly correct, he was unable to describe the origin of this force using the molecular theory of matter.

When the pedestrian bridge was being constructed, engineers observed that although the bridge appeared stable and safe for use, many residents, including learners, could not explain how the bridge remained in equilibrium. In particular, they were unable to identify the forces acting on the beam or determine the reaction exerted by the hinge.

To ensure long-term safety of the structure, the engineers decided to test the elastic properties of the steel wires used to reinforce the bridge. They intended to determine the Young's modulus of the wires before installation. However, the available technicians were unable to set up and explain the required experiment. The students were therefore requested to demonstrate the experiment using suitable laboratory apparatus and state the necessary precautions.

Hint

- Length of the beam = 4.0 m
- Weight of the beam = 50 N

Task

a) **Analyse** the information provided and explain, using the molecular theory of matter, the origin of the opposing force between the brake pads and the wheel rim that causes the wheel rim to become hot, identify all the forces acting on the bridge beam and determine the magnitude of the reaction at the hinge.
b) **Design the experimental** set up and explain to students to carry out the demonstration.

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SECTION B

Part (I)

ITEM 3

Due to the increasing threat of Ebola outbreaks in Uganda, many residents have become anxious whenever they experience minor symptoms such as headaches and fever. As a result, the number of people seeking medical examination at health facilities has increased significantly. However, several concerns have arisen among hospital workers, patients, and visitors.

In the hospital laboratory, a technician uses a **device that uses pressure of a fixed mass of the gas at a constant volume** to monitor the temperature of the room where medicines are stored. Some workers are concerned about how the device operates and how it can accurately measure temperature.

One of the admitted patients complained of a severe toothache. During examination, the dentist used a curved mirror to inspect the affected tooth before prescribing medication. The patient noticed that the image of the tooth appeared larger than normal and became curious about the type of mirror being used and how it produces such an image.

Meanwhile, students from a nearby school visited the hospital's science outreach centre during an astronomy exhibition. They observed the Moon using an optical instrument made of two convex lenses. The final image of the Moon was formed at infinity.

One student informed the others that a good viewing instrument should have a magnifying power of at least 6 to enable clear observation of distant objects. However, the students were unable to identify the type of instrument being used and derive the expression used to determine whether the instrument met the required standard. Their teacher later instructed them to prepare a written report based on their observations. Unfortunately, they were unable to complete the task and decided to seek assistance from a Senior Six Physics student from a nearby school.

Hint
Focal length of objective = 300 cm
Focal length of eyepiecee = 25 cm

Task

a) Analyse the concepts demonstrated and address the concerns raised by the hospital workers and the patient.
b) Analyze the information given and help the student with all information that is required in their report.

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ITEM 4

During the annual secondary school inter-class Music, Dance and Drama (MDD) competitions, the school band acquired several new guitars to improve their performance. However, during rehearsals, the house patrons noticed that some guitars were producing notes that were not properly in tune.

To address the issue, members of the Physics Club were requested to investigate and determine the frequencies of the notes produced by the guitar strings. Some students were unsure how this could be done and requested clarification, but the explanations given did not fully satisfy them, leaving the issue unresolved.

One guitar string of length **0.80 m** was selected for investigation. When plucked, the string vibrated in its **fundamental mode**, forming a stationary wave with a single loop between fixed ends. The frequency of vibration was measured to be **200 Hz**.

Further investigation raised additional concerns:

- Some students wanted to understand how stationary waves are formed on a stretched string.
- The mass per unit length of the string was found to be $\mathbf{2.0 \times 10^{-3}\ kg\ m^{-1}}$, but the students could not determine the tension in the string.
- Students were informed that well-tuned guitars operate at tensions of at least **204.8 N**, but they were unsure how to use this information to select suitable guitars.

At the same time, in the school canteen, the attendant was preparing a sugar solution for refreshments during the festival. However, there was uncertainty about whether the correct concentration of the sugar solution had been achieved. The attendant could not clearly explain how concentration can be determined, leading to concerns among the students.

TASK

(a) Analyse the information provided and provide the necessary information to help the students address the challenges arising from their investigation.

(b) Examine the canteen attendant concerns in preparing and measuring the concentration of sugar.

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Part (II)

ITEM 5

A rural district in Uganda has established a **Renewable Energy and Technical Innovation Centre** to improve access to electricity for schools, health centres, and small industries. The centre has installed electronic equipment, including solar power systems, communication devices, and a high-voltage laboratory for training technicians.

During routine maintenance, several challenges arose. The backup energy storage unit in one of the electronic control systems failed, and many trainee technicians could not explain how the electrical component responsible for storing electric charge is charged and discharged. They were also unable to sketch the graphs that describe the charging and discharging processes.

To improve energy efficiency, an engineer proposed storing electrical energy in a **200 μF energy-storage device** charged by a **12.0 V battery** and later using the stored energy to power a small electric heating coil immersed in **2 litres of water** for a laboratory demonstration. Before purchasing a new heating coil, the technicians needed to determine whether the energy stored in the device would produce the required increase in the water temperature. According to the manufacturer's specification, a suitable heating coil should produce a temperature rise of **at least** $1.71 \times 10^{-6}\,°C$ when connected to the charged device. However, the trainees were unable to perform the necessary calculations or determine whether the heating coil met the required specification.

The centre also plans to demonstrate the production of very high voltages using an **electrostatic machine to run the factory.** However, many workers could neither explain how the machine builds up high voltages nor illustrate its operation using a labelled diagram.

Hint

Specific heat capacity of water $= 4200 Jkg^{-1}K^{-1}$

Tasks

a) **Analyse** the concepts and address the challenges faced by the trainees.
b) **Examine the** information provided and explain whether the heating coil will meet the required specification and hence address the concerns of the workers.

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ITEM 6

John woke up very early in the morning to prepare for work. He wanted to iron his clothes, but the electricity supply was too low to operate the electric flat iron. Although some light bulbs were glowing, they were very dim, and several electrical appliances could not function properly. Wondering why this was happening, John contacted his son, who was studying Physics in Senior Six. His son suggested that a device capable of stepping up an alternating voltage to the required value should be installed. John became curious and asked his son to explain how the device works. He also requested his son to explain the most likely causes of energy losses in the device and how such losses can be minimized.

As John was about to leave for work, his car failed to start. He contacted a nearby mechanic who, after inspecting the vehicle, concluded that the battery's performance was too low. The battery was disconnected so that its maximum power output could be investigated. However, the mechanic could not derive the expression for the maximum power delivered by the battery in terms of its electromotive force (e.m.f.) and internal resistance. He also failed to sketch the graph of power output against load resistance.

To determine the battery's e.m.f., the mechanic connected an ammeter in series with the battery and a **2 Ω** resistor. The ammeter read **0.50 A**. When the **2 Ω** resistor was replaced with a **5 Ω** resistor, the ammeter reading dropped to **0.25 A**. However, the mechanic was unable to calculate the e.m.f. and determine whether the battery was suitable for starting the car. The minimum e.m.f. required for the battery to start the car is **1.5 V**.

Tasks

a) Design the device and help John understand the concerns he raised with his son.
b) Analyze the information and help the mechanic address all the challenges encountered during the battery test.

END

© 2026 Jinja Joint Examinations Board **End**